

Erdős Problem #467

EP-467

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Prove the following for all large x : there is a choice of congruence classes a_p for all primes $p \leq x$ and a decomposition $\{p \leq x\} = A \sqcup B$ into two non-empty sets such that, for all $n < x$, there exist some $p \in A$ and $q \in B$ such that $n \equiv a_p \pmod{p}$ and $n \equiv a_q \pmod{q}$.

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